

Please follow the instructions at this link:

<https://vectorbyteorg.github.io/bayesTPC-tutorials/intro.html>

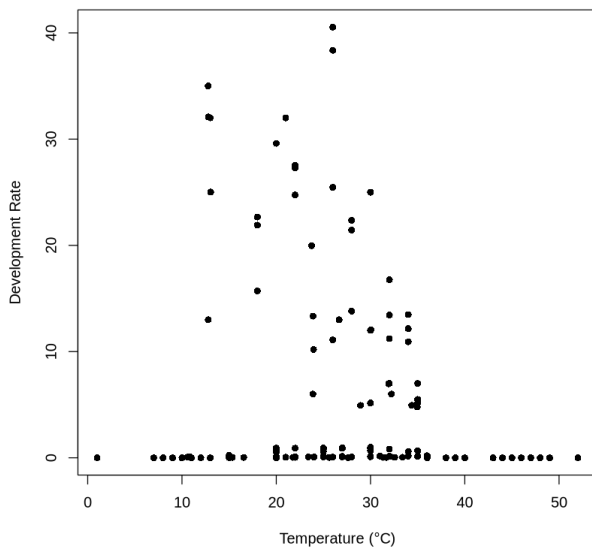
You can copy-paste the scripts below to run them.

```
In [2]: library(nimble)
library(bayesTPC)

Sys.setenv(PATH = paste("/opt/conda/envs/RGIS/bin",
                        Sys.getenv("PATH"), sep = ":"))
```

```
In [3]: # Load example data
data <- read.csv("data/AeaegyptiTraitData.csv")
data_list <- list(Trait = data$trait, Temp = data$Temp)

# Quick plot
plot(data_list$Trait ~ data_list$Temp, ylab = "Development Rate", xlab = "Temperature (°C)", pch = 16)
```



```
In [4]: set.seed(1)
fit <- b_TPC(data = data_list, model = "briere")
```

Warning message in check_data(data):
"Unusual (Temp>50) temperature values detected (are Temps given in Celcius?)"

[Creating NIMBLE model:](#)

- Configuring model.
- Compiling model.

[Creating MCMC:](#)

- Configuring MCMC.
- Compiling MCMC.
- Running MCMC.

[Progress:](#)

```
|-----|-----|-----|-----|
|-----|-----|-----|-----|
```

[Configuring Output:](#)

- Finding Max. a Post. parameters.

```
In [5]: print(fit)
```

[bayesTPC MCMC of Type:](#)

briere

[Formula:](#)

```
m[i] <- ( q * Temp * (Temp - T_min) * sqrt((T_max > Temp) * abs(T_max - Temp))
* (T_max > Temp) * (Temp > T_min) )
```

[Distribution:](#)

```
Trait[i] ~ T(dnorm(mean = m[i], tau = 1/sigma.sq), 0, )
```

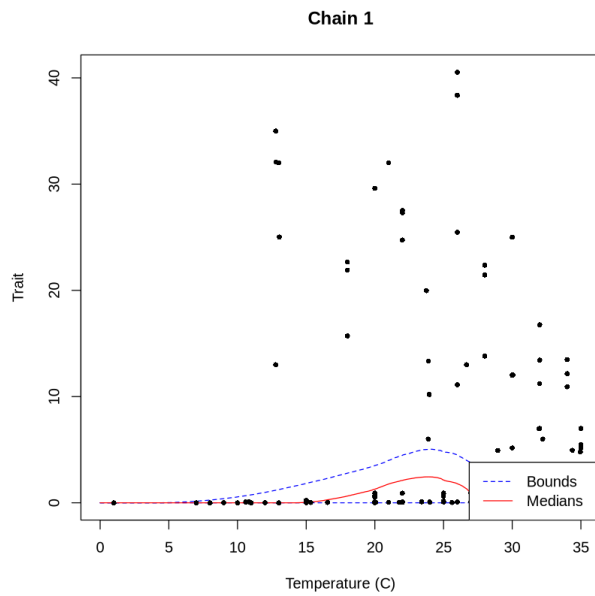
[Parameters:](#)

Chain 1:

	MAP	Mean	Median	Priors
T_max	25.016	28.948	28.192	dunif(25, 60)
T_min	19.977	12.828	14.216	dunif(0, 20)
q	0.057	0.010	0.005	dexp(1)
sigma.sq	60.460	61.131	61.175	dexp(1)

```
In [6]: plot(fit, temp_interval = seq(0, 35, length.out = 1000))
```

```
Warning message in plot.btpc_MCMC(fit, temp_interval = seq(0, 35, length.out = 1000)):  
"Using default title for plots. Either one title or individual titles for each plot may be provided."
```



```
In [ ]:
```